Mid Term Test
IT509 Data Mining and Data Warehousing
21 September, 2017
Marks: 40 Time: 1 hour

1. State true or false: $1 \times 5 = 5$
   a) Density based clustering techniques are capable of finding arbitrarily shaped clusters.
   b) The complete linkage-hierarchical algorithm finds compact but globular clusters.
   c) Computational complexity of a hierarchical algorithm is $O(n^2)$
   d) All patterns discovered by a data mining algorithm may not be interesting.
   e) A proximity measure means either a similarity measure or a dissimilarity measure.

2. a) How do you prepare a dataset for mining on it? 2
   b) What are the different types of attributes? Describe briefly. 4
   c) Name 3 categories of clustering algorithms along with important characteristics for each category. 4

3. a) What are the advantages of PAM over k-means? 2
   b) State two techniques for making PAM scalable. 2
   c) Present the CLARA algorithm. 4
   d) Access the computational complexity of CLARA. 2

4. a) Define the concept of density as applicable in DBSCAN. 2
   b) Describe the concept of density-rechability and density-connectivity. 4
   c) Utilize the two concepts stated in (b) above to provide the definition of a density-based cluster. 2
   d) What is a noise object? Show diagrammatically. 2

5. Gigen the following set of data points:

   (2, 3), (4, 8), (9, 2), (4, 2), (5, 5), (8, 5), (9, 10)

   a) Find ε-neighbourhood for point (5, 5) for ε = 4. 3
   b) Is (5, 5) a core point? 1
   c) Determine whether the point (9, 10) is directly density reachable from the point (5, 5). 1

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